# PG No. 22 $C_{3h}$ $\bar{6}$ [ hexagonal ]

* generator : $3^{+}_{001}$, $\mathrm{m}_{001}$
* conjugacy class

$$
\begin{aligned}
[1]: &\quad 1 \\
[3^{+}_{001}]: &\quad 3^{+}_{001},\ 3^{-}_{001} \\
[\mathrm{m}_{001}]: &\quad \mathrm{m}_{001} \\
[-6^{+}_{001}]: &\quad -6^{+}_{001},\ -6^{-}_{001}
\end{aligned}
$$

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

| No. | tag | matrix (polar) | det |
|---|---|---|---|
| 1 | 1 | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 2 | $3^{+}_{001}$ | $\begin{bmatrix} 0 & -1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 3 | $3^{-}_{001}$ | $\begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 4 | $\mathrm{m}_{001}$ | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ | $-1$ |
| 5 | $-6^{-}_{001}$ | $\begin{bmatrix} 0 & -1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ | $-1$ |
| 6 | $-6^{+}_{001}$ | $\begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ | $-1$ |